EVERY INTEGRAL CLOSED SUBSPACE HAS A UNIQUE DOMINANT IMAGE, PROPER AND SURJECTIVE WHEN THE MORPHISM IS PROPER

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a construction of the image through the generic point with a reduction argument for the morphism, and a short properness and surjectivity chain in the proper case

Editorial note. The label lemma-proper-image names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Integral closed subspaces and their generic points, the closure of the image of a generic point being the generic point of an integral closed subspace, and the reduction of a closed subspace
- The cartesian description of the points of the preimage of a closed subspace, and the universal property of the reduction
- Stability of properness under composition, under base change and for closed immersions, and closedness of the image of a proper morphism

Source reference. Original reference 0EDV: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 311–311. Prerequisite pointer invoked at proof line(s) 838; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02QL: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 264–264. Prerequisite pointer invoked at proof line(s) 821; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04WP: sources/stacks-chow/tags.txt, source lines 6367–6367. Prerequisite pointer invoked at proof line(s) 848; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04XZ: sources/stacks-chow/tags.txt, source lines 6412–6412. Prerequisite pointer invoked at proof line(s) 849; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 04NX: sources/stacks-chow/tags.txt, source lines 6095–6095. Prerequisite pointer invoked at proof line(s) 852; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03JJ: sources/stacks-chow/tags.txt, source lines 4641–4641. Prerequisite pointer invoked at proof line(s) 842; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03H4: sources/stacks-chow/tags.txt, source lines 4553–4553. Prerequisite pointer invoked at proof line(s) 839; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 02QL let $X, Y/B$ be good and let $f : X \to Y$ be a morphism over B. If $Z \subset X$ is an integral closed subspace, then there exists a unique integral closed subspace $Z' \subset Y$ such that there is a commutative diagram*

$$\begin{array}{ccc} Z & \longrightarrow & X \\ \downarrow & & \downarrow f \\ Z' & \longrightarrow & Y \end{array}$$

with $Z \to Z'$ dominant. If f is proper, then $Z \to Z'$ is proper and surjective.

Proof. Let $\xi \in |Z|$ be the generic point. Let $Z' \subset Y$ be the integral closed subspace whose generic point is $\xi' = f(\xi)$, see Remark 0EDV. Since $\xi \in |f^{-1}(Z')| = |f|^{-1}(|Z'|)$ by Properties of Spaces, Lemma 03H4 and since Z is the reduced with $|Z| = \overline{\{\xi\}}$ we see that $Z \subset f^{-1}(Z')$ as closed subspaces of X (see Properties of Spaces, Lemma 03JJ). Thus we obtain our morphism $Z \to Z'$. This morphism is dominant as the generic point of Z maps to the generic point of Z'. Uniqueness of Z' is clear. If f is proper, then $Z \to Y$ is proper as a composition of proper morphisms (Morphisms of Spaces, Lemmas 04WP and 04XZ). Then we conclude that $Z \to Z'$ is proper by Morphisms of Spaces, Lemma 04NX. Surjectivity then follows as the image of a proper morphism is closed. □